

Problem #2

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Due: Tuesday, October 28, turn in to canvas, as described in the syllabus.

1. Determine for which values of $\beta > 0$, there is “finite-time blow-up” of solutions (i.e. there exists a finite $T > 0$ such that $\lim_{t \rightarrow T} |x(t)| = +\infty$) for the following scalar ode:

$$\dot{x} = x^\beta, \quad x(0) = x_0 \in \mathbb{R}_+.$$

Specify the domain of existence of the solution, i.e. determine T as a function of x_0 and β .

In the remainder of this problem, let's focus on $\beta = 2$ and $x_0 = 1$. What is the time T to “blow up” in this case? Introduce the following transformation from t to a new time variable τ :

$$\tau = \int_0^t 1 + x^2(s) ds, \quad t \in [0, T)$$

where $x(s)$ satisfies the original problem $dx/ds = x^2$. Show that this is a one-to-one mapping between $t \in [0, T)$ and τ , and specify the range of τ . Let $y(\tau) = x(t(\tau))$; what differential equation does $y(\tau)$ satisfy? Does $y(\tau)$ have finite time blow-up? (You may find it helpful to read Section 4.3 of the textbook on Global Existence of Solutions.)

Solution. (i) We have $\frac{dx}{dt} = x^\beta$. Then

$$\begin{aligned} x^{-\beta} dx &= dt \\ \int_{x_0}^{x(t)} x^{-\beta} dx &= \int_0^t ds \\ \frac{x^{-\beta+1}}{-\beta+1} \Big|_{x_0}^{x(t)} &= t \quad (\beta \neq 1), \quad \ln |x| \Big|_{x_0}^{x(t)} = t \quad (\beta = 1) \\ x(t) &= ((-\beta+1)t + x_0^{-\beta+1})^{\frac{1}{-\beta+1}} \quad (\beta \neq 1), \quad x(t) = e^t x_0 \quad (\beta = 1) \end{aligned}$$

Clearly for $\beta = 1$, $x(t)$ does not blow-up in finite time. For $\beta < 1$, $x(t)$ is well defined for all $t > 0$. For $\beta > 1$, $x(t)$ blows up when $(-\beta+1)t + x_0^{-\beta+1} = 0$, i.e. $T = \frac{x_0^{-\beta+1}}{\beta-1}$.

(ii) Let $\beta = 2$ and $x_0 = 1$. Then $T = 1$, $x(t) = \frac{1}{1-t}$. We have

$$\tau = \int_0^t 1 + x^2(s) ds = \int_0^t 1 + \frac{1}{(1-s)^2} ds = t + \frac{1}{1-t} - 1$$

We compute $\frac{d\tau}{dt} = 1 + x^2(t) = 1 + \frac{1}{(1-t)^2} > 0$ for $t \in [0, 1)$. Hence $\tau(t)$ is monotone increasing and the map is one-to-one. $\tau(0) = 0$ and $\tau(1) = \infty$. Hence the range of τ is $[0, \infty)$.

Let $y(\tau) = x(t(\tau))$. We have $\dot{y} = \frac{d}{d\tau}y(\tau) = \frac{d}{d\tau}x(t(\tau)) = \frac{dx}{dt} \frac{dt}{d\tau} = x^2 \frac{1}{1+y^2} = \frac{y^2}{1+y^2}$. Since $\dot{y} < 1$, it is bounded by $y(0) + \tau$. Hence it does not blow up in finite time. ▲

2. Consider the scalar initial value problem

$$\dot{x} = -|x|^\beta, \quad x(0) = 0.$$

For which values of $\beta > 0$ is the solution $x(t) = 0$ guaranteed to be unique on some interval $J = [-a, a]$, $a > 0$? If there is no such guarantee, explicitly construct a family of solutions to the initial value problem on the interval J .

Solution. By the existence and uniqueness theorem of ODE, $x(t) = 0$ will have unique solutions if $-|x|^\beta$ is locally Lipschitz. For $\beta \geq 1$, $-|x|^\beta$ is locally Lipschitz around $x = 0$. Indeed, let $V(x) = -|x|^\beta$. If $\beta = 1$, $V(x) = -|x|$ and $|V(x) - V(y)| = | -|x| + |y| | = ||y| - |x|| \leq |x - y|$, for all x, y . Hence $V(x)$ is Lipschitz for $\beta = 1$.

For $\beta > 1$, let $g : [-r, r] \rightarrow \mathbb{R}$ such that $g(x) = x^\beta$, then $g'(x) = \beta x^{\beta-1}$. Then, by mean value theorem, there exists z such that $|g(x) - g(y)| = |g'(z)||x - y|$. Clearly $|g'(z)| = \beta|z|^{\beta-1} < \beta r^{\beta-1}$. Hence $V(x)$ is Lipschitz on $(-r, r)$ for some $r > 0$.

For $\beta < 1$, $-|x|^\beta$ is not Lipschitz. For any $r > 0$ and $L > 0$, pick $x \in [-r, r]$, $\frac{|V(x) - V(0)|}{|x - 0|} = |x|^{\beta-1}$, as $\beta - 1 < 0$, by letting x arbitrary small we can have $|x|^{\beta-1} \geq L$. Hence $V(x)$ is not Lipschitz.

Clearly we have $x_1(t) = 0$, $t \in J$ as the trivial solution. Then, let $0 \leq \tau < a$ where $x(t) = 0$ for $0 \leq t \leq \tau$. For $\tau < t \leq a$, we solve

$$\begin{aligned} \frac{dx}{dt} &= -|x|^\beta \\ \int_0^{x(t)} -|x|^{-\beta} dx &= \int_\tau^t ds \\ \begin{cases} \left. \frac{-x^{-\beta+1}}{-\beta+1} \right|_0^{x(t)} &= t - \tau, \quad x > 0 \\ \left. \frac{-x^{-\beta+1}}{-\beta+1} \right|_0^{-x(t)} &= t - \tau, \quad x < 0 \end{cases} \\ x(t) &= -((-\beta + 1)(t - \tau))^{\frac{1}{-\beta+1}} \end{aligned}$$

Clearly, $x(t) < 0$ for $t - \tau > 0$. Hence,

$$x_2(t) = \begin{cases} 0, & 0 \leq t \leq \tau \\ -((-\beta + 1)(t - \tau))^{\frac{1}{-\beta+1}}, & \tau < t < a \end{cases}$$

Now, for $t < 0$, let $-a < \tau' \leq 0$ such that $x(t) = 0$ for $t \geq \tau'$. Then for $-a \leq t < \tau'$ we solve

$$\begin{aligned} \frac{dx}{dt} &= -|x|^\beta \\ \int_0^{x(t)} -|x|^{-\beta} dx &= \int_{\tau'}^t ds \\ \begin{cases} \left. \frac{-x^{-\beta+1}}{-\beta+1} \right|_0^{x(t)} &= t - \tau', & x > 0 \\ \left. \frac{-x^{-\beta+1}}{-\beta+1} \right|_0^{-x(t)} &= t - \tau', & x < 0 \end{cases} \\ x(t) &= -((- \beta + 1)(t - \tau'))^{\frac{1}{-\beta+1}} \end{aligned}$$

Clearly, $x(t) > 0$ for $t - \tau' < 0$. Hence,

$$x_3(t) = \begin{cases} 0, & \tau' \leq t \leq 0 \\ ((-\beta + 1)(\tau' - t))^{\frac{1}{-\beta+1}}, & -a \leq t < \tau' \end{cases}$$

Since τ and τ' are arbitrary, we have an infinite family of solutions on J for $\beta < 1$. ▲

3. Often in applications, the dynamics of a system being modeled, may be governed by quite different processes in different regions of the state space. And this might lead to a dynamical system that is set-wise defined, e.g.

$$\dot{\mathbf{x}} = \begin{cases} \mathbf{f}^+(\mathbf{x}), & \sigma(\mathbf{x}) > 0, \\ \mathbf{f}^-(\mathbf{x}), & \sigma(\mathbf{x}) < 0, \end{cases} \quad (1)$$

where the scalar equation $\sigma(\mathbf{x}) = 0$ defines the boundary between different regions of state space governed by different dynamics. (Note that I didn't say anything about the dynamics for points **in** the boundary.)

In the following, you explore how various possible definitions of the dynamics in the boundary might impact the behavior of the system. Specifically, consider (1) with $\mathbf{f}^\pm(\mathbf{x})$ given as a planar, linear vector field

$$\mathbf{f}^\pm(\mathbf{x}) = \begin{pmatrix} -1 \pm x_2 \\ x_2 \mp x_1 \end{pmatrix},$$

and let the boundary be defined by $\sigma(\mathbf{x}) = x_2$.

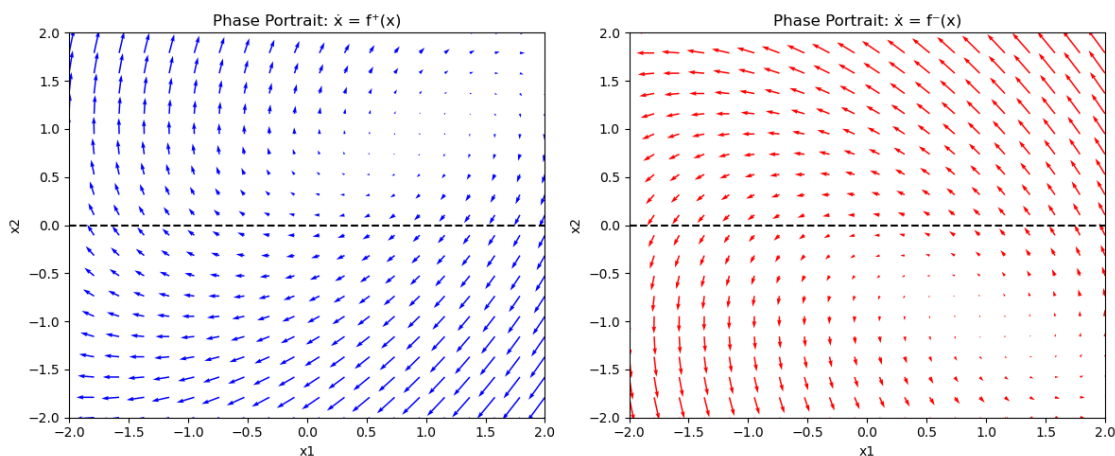
- Plot the phase portrait associated with $\dot{\mathbf{x}} = \mathbf{f}^+(\mathbf{x})$ and $\dot{\mathbf{x}} = \mathbf{f}^-(\mathbf{x})$, separately in the square region $x_1, x_2 \in [-2, 2]$, i.e. don't restrict them to their regions $x_2 > 0$ or $x_2 < 0$.
- Show that the vector field is discontinuous on $x_2 = 0$.
- Let's proceed by restricting to consideration of the forward time behavior of the model with some prescribed dynamics on the discontinuity boundary. For example, one natural way that the dynamics might be defined is as some convex combination of $\mathbf{f}^+(\mathbf{x})$ and $\mathbf{f}^-(\mathbf{x})$, i.e. let

$$\dot{\mathbf{x}} = \alpha \mathbf{f}^+(\mathbf{x}) + (1 - \alpha) \mathbf{f}^-(\mathbf{x}), \quad \text{for } \mathbf{x} \text{ satisfying } \sigma(\mathbf{x}) = 0,$$

where $\alpha \in [0, 1]$ is a parameter. For the following choices of α , plot the forward solution in the phase plane for an initial condition $(x_1, x_2) = (2, 1)$. (This can be done as a somewhat qualitative sketch.)

(i) $\alpha = 0$, (ii) $\alpha = 0.5$, (iii) $\alpha = 1$. Comment on whether the behavior is sensitive to the choice of the parameter $\alpha \in [0, 1]$.

Solution. (a) See plot attached:

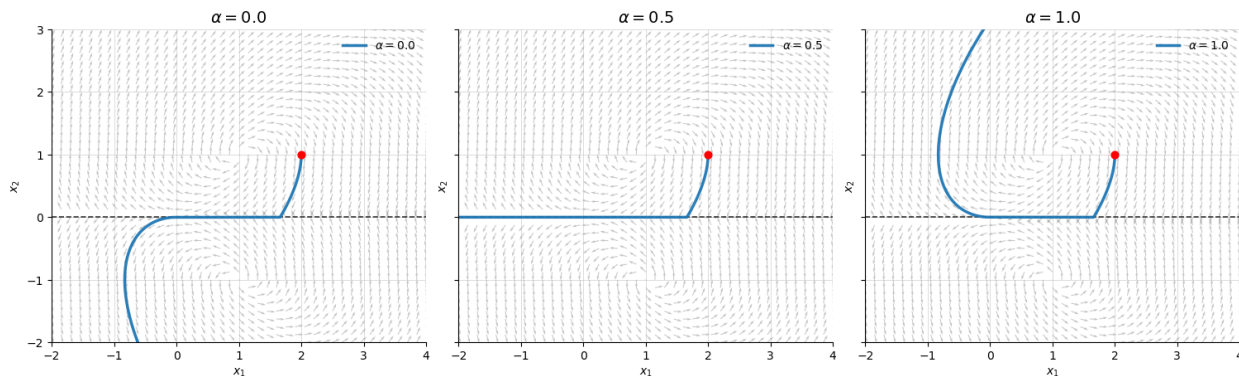


(b) We have

$$\dot{\mathbf{x}} = V(\mathbf{x}) = \begin{cases} \begin{pmatrix} -1 + x_2 \\ x_2 - x_1 \end{pmatrix}, & x_2 > 0 \\ \begin{pmatrix} -1 - x_2 \\ x_2 + x_1 \end{pmatrix}, & x_2 < 0 \end{cases}$$

$\lim_{x_2 \rightarrow 0^-} V(\mathbf{x}) = \begin{pmatrix} -1 \\ x_1 \end{pmatrix} \neq \begin{pmatrix} -1 \\ -x_1 \end{pmatrix} = \lim_{x_2 \rightarrow 0^+} V(\mathbf{x})$. Hence the vector field is discontinuous on $x_2 = 0$.

(c) See plot attached



At the initial condition, we have the trajectory moving with direction $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$. Then the

trajectory is moving down at first. We have $f_\alpha(x_1, 0) = \begin{pmatrix} -\alpha \\ -\alpha x_1 \end{pmatrix} + \begin{pmatrix} \alpha - 1 \\ (1 - \alpha)x_1 \end{pmatrix} = \begin{pmatrix} -1 \\ (1 - 2\alpha)x_1 \end{pmatrix}$. All three trajectories are the same until the first contact with the discontinuity $x_2 = 0$. Then, x_2 is attracted to 0 as $\mathbf{f}^+(x_1, 0) = \begin{pmatrix} -1 \\ -x_1 \end{pmatrix} = \mathbf{f}^-(x_1, 0)$, and x_1 is always decreasing when $x_2 = 0$. This happens until x_1 becomes negative. Then, for $\alpha = 0$, the vertical component at the boundary is $x_1 < 0$. Hence the trajectory will go downwards. For $\alpha = 0.5$, the vertical component is 0. Hence the trajectory continues to stay at $x_2 = 0$. For $\alpha = 1$, the vertical component at the boundary is $-x_1 > 0$. Hence the trajectory will go upwards. The behavior of the trajectory is sensitive to the choice of the parameter α only when it x_1 becomes negative. ▲

4. Use Hartman-Grobman theorem to prove that linear asymptotic stability implies asymptotic stability for an equilibrium $\mathbf{x}^*$ of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, where $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^1 vector field. Can you also use it to show that linear instability implies instability?

Solution. Let $A = \nabla \mathbf{f}(\mathbf{x})$. By linear asymptotic stability, $\Re(\lambda) < 0$ for any eigenvalues λ of A , which implies the equilibrium $\mathbf{x}^*$ is hyperbolic. By Hartman-Grobman theorem, there exists neighborhood $U \ni \mathbf{x}^*$ and a homeomorphism $h : V \rightarrow U$ such that $\varphi_t = h \circ e^{tA} \circ h^{-1}$ for all $t \in \mathbb{R}$, where V is a neighborhood around 0. Hence we have $\varphi_t(h(y)) = h(e^{tA}(y))$ for all $y \in V$. Then, $\varphi_t(h(0)) = h(0)$, and hence $h(0) = \mathbf{x}^*$. Since h is a homeomorphism, for every $z \in U$ there exists $y \in V$ such that $z = h(y)$. By linear stability, $\lim_{t \rightarrow \infty} e^{tA}y = 0$. Hence, $\lim_{t \rightarrow \infty} \varphi_t(z) = \lim_{t \rightarrow \infty} \varphi_t(h(y)) = \lim_{t \rightarrow \infty} h(e^{tA}y) = h(0) = \mathbf{x}^*$. It remains to show that $\mathbf{x}^*$ is Lyapunov stable. Indeed, the linear system is Lyapunov stable. Let $\epsilon > 0$. By continuity of h , there exists $\delta > 0$ such that if $\|y\| < \delta$, then $\|h(y) - h(0)\| = \|h(y) - \mathbf{x}^*\| < \epsilon$. By the stability of the linear system, for y in a η -neighborhood of 0, i.e., there exists η such that if $\|y\| < \eta$, then $\|e^{tA}y\| < \delta$, i.e., it is within the δ -neighborhood. Since h is a homeomorphism, h^{-1} is continuous. Hence, there exists $\rho > 0$ such that if $\|x - \mathbf{x}^*\| < \rho$, then $\|h^{-1}(x) - h^{-1}(\mathbf{x}^*)\| = \|h^{-1}(x)\| < \eta$. Combining them together, pick x_0 such that $\|x_0 - \mathbf{x}^*\| < \rho$, then $\|h^{-1}(x_0)\| < \eta$. Set $y_0 = h^{-1}x_0$, then $\|y_0\| < \eta$ implies $\|e^{tA}y_0\| < \delta$. By continuity of h , $\|\varphi_t(x_0) - \mathbf{x}^*\| = \|h(e^{tA}(y_0)) - \mathbf{x}^*\| = \|h(e^{tA}(y_0)) - h(0)\| < \epsilon$. Hence $\mathbf{x}^*$ is Lyapunov stable. Therefore, linear asymptotic stability implies asymptotic stability.

For linear instability, we have $\Re(\lambda) \geq 0$ for some eigenvalues λ of A . Then the linear system is not Lyapunov stable, i.e., there exists a ϵ -neighborhood around 0 such that for all δ -neighborhood, there exists some $\{y_k\}_{k=1}^\infty$ converges to 0 and times $\{t_k\}_{k=1}^\infty$ with $\|y_k\| < \delta$, we have $\|e^{t_k A}y_k\| \geq \epsilon$. By continuity of h^{-1} , for any ϵ , there exists η such that if $\|x - \mathbf{x}^*\| < \eta$, then $\|h^{-1}(x) - h^{-1}(\mathbf{x}^*)\| = \|h^{-1}(x)\| < \epsilon$. Let $x_k = h(y_k)$, then for every k . With $\|y_k\| = \|h^{-1}(x_k)\| \geq \epsilon$, we have $\|x_k - \mathbf{x}^*\| = \|h(y_k) - \mathbf{x}^*\| \geq \eta$. By continuity of h , we have $\lim_{k \rightarrow \infty} x_k = \lim_{k \rightarrow \infty} h(y_k) = h(0) = \mathbf{x}^*$. By Hartman-Grobman, since $\|e^{t_k A}y_k\| \geq \epsilon$, $\|\varphi_{t_k}(x) - \mathbf{x}^*\| = \|h(e^{t_k A}y_k) - \mathbf{x}^*\| \geq \eta$. Hence $\mathbf{x}^*$ incurs instability. ▲

5. In Section 4.8 of the textbook, on the Hartman Grobman theorem, there is an example of constructing a homeomorphism $H(x, y) = (x, y - x^2/3)$ that satisfies the topological conjugacy

condition $\psi_t \circ H = H \circ \phi_t$ where ϕ_t is generated by

$$\begin{aligned}\dot{x} &= x \\ \dot{y} &= -y + x^2.\end{aligned}$$

Solve these differential equations to obtain ϕ_t and ψ_t , where ψ_t applies to the linearized flow. Then check, by explicit computation, that indeed $\psi_t \circ H = H \circ \phi_t$.

Solution. Denote the system $\dot{\mathbf{x}} = V(\mathbf{x}) = V(x, y)$. We notice that the equilibrium point of this system is $(0, 0)$ as $V(0, 0) = 0$. The linearization of this system at $(0, 0)$ is

$$A = \nabla V(0, 0) = \begin{pmatrix} \partial_x x & \partial_y x \\ \partial_x(-y + x^2) & \partial_y(-y + x^2) \end{pmatrix} \Big|_{(x,y)=(0,0)} = \begin{pmatrix} 1 & 0 \\ 2x & -1 \end{pmatrix} \Big|_{(x,y)=(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Since this linear system is diagonal, we have $\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{At} \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} e^t x(0) \\ e^{-t} y(0) \end{pmatrix}$. For the solutions of the nonlinear system, solving $\dot{x} = x$ yields $x(t) = e^t x(0)$. Then we have $\dot{y} = -y + e^{2t} x(0)^2$. Equivalently, we have $e^t \dot{y} + e^t y = e^{3t} x(0)^2$, i.e., $\frac{d}{dt}(e^t y) = e^{3t} x(0)^2$. Hence, we have $e^t y(t) = \frac{e^{3t}}{3} x(0)^2 + C$, then $y(0) = C + \frac{1}{3} x(0)^2$. Hence, $y(t) = e^{-t} y(0) + \frac{1}{3} (e^{2t} - e^{-t}) x(0)^2$. Hence, we have

$$\psi_t(x, y) = \begin{pmatrix} e^t x \\ e^{-t} y \end{pmatrix}, \quad \phi_t(x, y) = \begin{pmatrix} e^t x \\ e^{-t} y + \frac{1}{3} (e^{2t} - e^{-t}) x^2 \end{pmatrix}.$$

We have $H(x, y) = (x, y - x^2/3)$. Then

$$\psi_t \circ H(x, y) = \begin{pmatrix} e^t x \\ e^{-t} (y - \frac{x^2}{3}) \end{pmatrix}.$$

Then,

$$H \circ \phi_t(x, y) = \begin{pmatrix} e^t x \\ e^{-t} y + \frac{1}{3} (e^{2t} - e^{-t}) x^2 - \frac{(e^t x)^2}{3} \end{pmatrix} = \begin{pmatrix} e^t x \\ e^{-t} (y - \frac{x^2}{3}) \end{pmatrix}.$$

Hence, $\psi_t \circ H = H \circ \phi_t$. ▲

6. Consider the Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz\end{aligned}$$

where r, b, σ are positive constants and $(x, y, z) \in \mathbb{R}^3$.

- (a) Show that the origin is the only fixed point for $r \in (0, 1]$, and that the following is a strong Lyapunov function for it for $r < 1$:

$$L = \frac{1}{2} \left(\frac{x^2}{\sigma} + y^2 + z^2 \right).$$

- (b) Show that L is only a weak Lyapunov function if $r = 1$, but that you can use LaSalle's Invariance Principle to prove that the origin is asymptotically stable for $r = 1$ as well as for $r < 1$. Note that you did not have to restrict any of this to a particular neighborhood U of the origin. The origin is globally attracting for the Lorenz equations when $r \in (0, 1]$.

Solution. (a) We need to solve this system of equations:
$$\begin{cases} \sigma(y - x) = 0 \\ rx - y - xz = 0 \\ xy - bz = 0 \end{cases} \quad \text{Indeed } (0, 0, 0)$$

is a solution for this system of equations. To show that there are no other solutions for $r \in (0, 1]$, from the first equation we have $x = y$, and from the third equation we have $z = x^2/b = y^2/b$. From the second equation we have $rx - x - \frac{x^3}{b} = 0$. Since $x = 0$ immediately gives $y = 0$ and $z = 0$, we assume $x \neq 0$ and divide both sides by x and we get $-\frac{x^2}{b} + (r - 1) = 0$. Since $r - 1 \leq 0$, we must have $-\frac{x^2}{b} \geq 0$, which gives the only possible solution $x = 0$. Hence, there is no other fixed point different from the origin.

Denote the system as $\dot{\mathbf{x}} = V(\mathbf{x})$, where $\mathbf{x} = (x, y, z)$. Let $L = \frac{1}{2} \left(\frac{x^2}{\sigma} + y^2 + z^2 \right)$. It is easy to check that $L(0, 0, 0) = 0$ and $L(x, y, z) > 0$ for $(x, y, z) \neq (0, 0, 0)$. To show the last condition, we check that $\frac{dL}{dt} < 0$. Indeed,

$$\begin{aligned} \frac{dL}{dt} &= \nabla L(\mathbf{x}) \cdot \dot{\mathbf{x}} = \nabla L(x, y, z) \cdot V(x, y, z) \\ &= \begin{pmatrix} \frac{x}{\sigma} \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} \sigma(y - x) \\ rx - y - xz \\ xy - bz \end{pmatrix} = yx - x^2 + rxy - y^2 - xyz + xyz - bz^2 \\ &= (1 + r)xy - x^2 - y^2 - bz^2 \\ &= -(x^2 - (1 + r)xy + \frac{(1 + r)^2}{4}y^2) - \frac{4 - (1 + r)^2}{4}y^2 - bz^2 \\ &= -(x - \frac{1 + r}{2}y)^2 - \frac{(1 - r)(r + 3)}{4}y^2 - bz^2 \end{aligned}$$

For $r < 1$, $(1 - r)(r + 3) > 0$, and thus $\frac{dL}{dt} < 0$ for any $(x, y, z) \neq (0, 0, 0)$. Since $\frac{dL}{dt} < 0$ for all t and for all $\mathbf{x} \neq \mathbf{x}^*$, integrating both sides yields $L(\mathbf{x}(t)) - L(\mathbf{x}(0)) = \int_0^t \nabla L(\mathbf{x}(s)) \cdot V(\mathbf{x}(s)) dt < 0$, which implies $L(\mathbf{x}(t)) - L(\mathbf{x}(0)) < 0$, i.e. for any $\mathbf{x} \neq \mathbf{x}^*$, $L(\varphi_t(\mathbf{x})) - L(\mathbf{x}) < 0$. Hence, L is a strong Lyapunov function for $r < 1$.

- (b) If $r = 1$, we have $\frac{dL}{dt} = -(x - y)^2 - bz^2 \leq 0$ for every (x, y, z) . Then the above results changes to for any $\mathbf{x}$, $L(\varphi_t(\mathbf{x})) - L(\mathbf{x}) \leq 0$. Hence L is only a weak Lyapunov function globally for $r = 1$. Let $c > 0$. Let U be the level sets of L , i.e., $U = \{\mathbf{x} : L(\mathbf{x}) \leq c\}$. Clearly such set is bounded as $\lim_{\|\mathbf{x}\| \rightarrow \infty} L(\mathbf{x}) = \infty$, and is closed by continuity, and hence is compact (since the setting is finite-dimensional). It is also forward invariance since $L(\varphi_t(\mathbf{x})) \leq L(\mathbf{x}) \leq c$ by the properties of Lyapunov function. Let $Z = \{\mathbf{x} \in U : \frac{dL}{dt} = 0\}$, then we have $-(x - y)^2 - bz^2 = 0$ for $(x, y, z) \in Z$, i.e. $x = y$, $z = 0$. In this set, $\dot{x} = 0$, $\dot{y} = 0$, $\dot{z} = x^2 = y^2$. Hence, to remain in this set, $\dot{z} = 0$, which implies $x = y = 0$. Hence the largest forward invariance subset of Z is $\{\mathbf{x}^*\}$. Then by LaSalle's Invariance Principle, we can conclude that $\mathbf{x}^* = (0, 0, 0)$ is asymptotically stable and attracts every point in U for $r = 1$. Since c is arbitrary, we can construct U such that it contains any given point in $\mathbb{R}^3$. Hence it attracts every point globally. For $r < 1$, L is a strong Lyapunov function globally, which implies $\mathbf{x}^* = (0, 0, 0)$ is asymptotically stable.

